

UMER NAWAZ

AI / ML Engineer · Generative AI · LLMs, RAG & Agentic Systems

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SUMMARY

Computer Science undergraduate at UET Lahore specializing in applied AI — large language models, retrieval-augmented generation, and agentic systems. Built end-to-end AI applications spanning document ingestion, vector search, speech transcription, multi-agent orchestration, and tool-calling workflows, on a strong foundation in Python, machine learning, and backend engineering. Ready to contribute to software and AI engineering teams by developing scalable, intelligent, and impactful solutions.

EDUCATION

University of Engineering and Technology (UET), Lahore

Bachelor of Science in Computer Science

- Relevant coursework: Data Structures & Algorithms, OOP, Database Systems, Operating Systems, Machine Learning, AI, NLP.

TECHNICAL SKILLS

Programming: Python, C++, SQL

Generative AI: LLM integration, Retrieval-Augmented Generation (RAG), LangChain, AI agents & agentic workflows, tool calling, prompt engineering, embeddings, semantic search

Machine Learning: Scikit-learn, NLP (spaCy, Sentence Transformers, Whisper), deep learning fundamentals, feature engineering, model evaluation, Pandas, NumPy

Backend & Frameworks: FastAPI, Django, Streamlit

Data & Storage: PostgreSQL, SQL Server, ChromaDB (vector database)

Tools & Platforms: Git, GitHub, Docker, Linux, VS Code, Code block

PROJECTS

Student Assistant — AI-Powered RAG Chatbot

Python, ChromaDB, Streamlit

- Built an end-to-end retrieval-augmented question-answering assistant that lets students query their own lecture notes and PDFs in natural language.
- Engineered the full ingestion pipeline: PDF text extraction, recursive chunking, Sentence-Transformer embeddings, and persistence in a ChromaDB vector store.
- Ranked retrieved chunks with Maximal Marginal Relevance (MMR) to cut redundant context, and grounded LLM answers strictly in retrieved passages to limit hallucination.

Video-Agent — Meeting Intelligence System

Whisper, FFmpeg, ChromaDB

- Built a pipeline that converts meeting recordings and video links into structured, searchable intelligence: media acquisition and audio extraction with yt-dlp and FFmpeg, then speech-to-text with OpenAI Whisper.
- Generated transcripts, translations, summaries, key points, decisions, open questions, and action items from raw audio, with language-aware handling for English and Urdu.
- Indexed long transcripts in ChromaDB so users can retrieve specific moments and follow-ups from hours of recordings.

DeepSearch — Multi-Agent Web Research System

Agentic AI, Python

- Designed a multi-agent system that decomposes a research question into specialized subtasks handled by coordinated agents, with web scraping and content-extraction workflows to gather and clean source material.
- Orchestrated agent hand-offs so partial findings are synthesized into a single consolidated, source-backed research report.

Real-Time Weather & News Agent

Tool Calling, LLM APIs

- Built a tool-calling agent that reasons about a request, selects the right external API, and merges live weather and news data into one response.
- Applied agentic patterns — tool schemas, action selection, result synthesis — instead of static prompt chaining.

NLP Applications — Grammar / Topic Analyzer & Emotion Detector

spaCy, Scikit-learn

- Built a spaCy analyzer that turns unstructured text into structured linguistic insight (grammar and topic extraction) behind an interactive UI.
- Trained and evaluated a text-classification model that detects emotional tone in user input.

CERTIFICATIONS & LANGUAGES

- Claude with Anthropic 101** and **Claude AI Fluency for Students** — Anthropic
- Artificial Intelligence & Machine Learning** — NAVTTC (National Vocational & Technical Training Commission)
- Languages:** Urdu (Native), English (Professional working proficiency)